

Power Factor Corrector



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This project discusses the need for power factor correction and provides a suitable DIY solution that could be used for small-scale industries and establishments.

When voltage and current are in phase with each other in an AC circuit, the electrical energy drawn from the source gets fully converted into another form of energy and the power factor (cosine of angle between voltage and current waveform) is said to be unity (or 100 per cent). This happens with purely resistive loads. With inductive loads, the voltage and current do not remain in phase and the power factor drops.

As the power factor drops, the system becomes less efficient. A drop from unity to 0.9 (90 per cent) in the power factor results in 15 per cent more current requirement for the same load. A power factor of 0.7 (70 per cent) requires approximately 43 per cent more current.

In industrial units and establishments, most of the loads are electrical motors and air-conditioning units. These loads are inductive in nature, where the current lags the applied voltage and the power factor is termed as lagging power factor. With capacitive loads, the current leads the voltage and the power factor is termed as leading power factor. The objective therefore should be to neutralise the lagging power factor of inductive loads by connecting capacitors across the load, which have leading power factor.

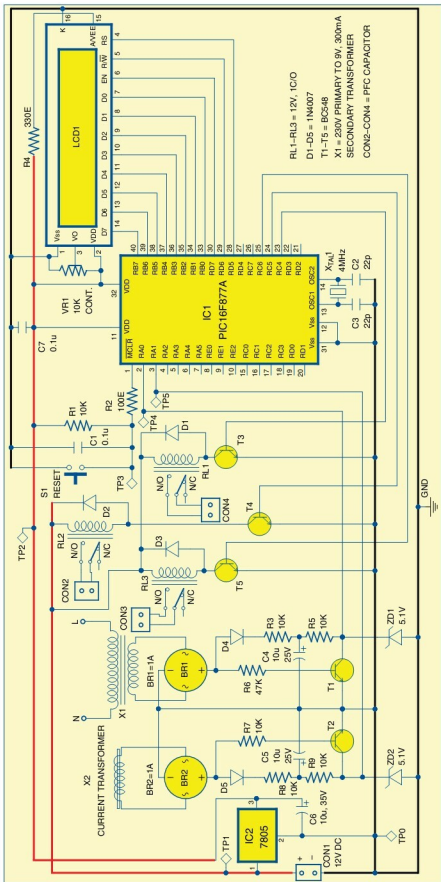


Fig. 1: The power factor correction circuit